

NEUROTECH BEDSIDE

Hackathon 2026 Submission

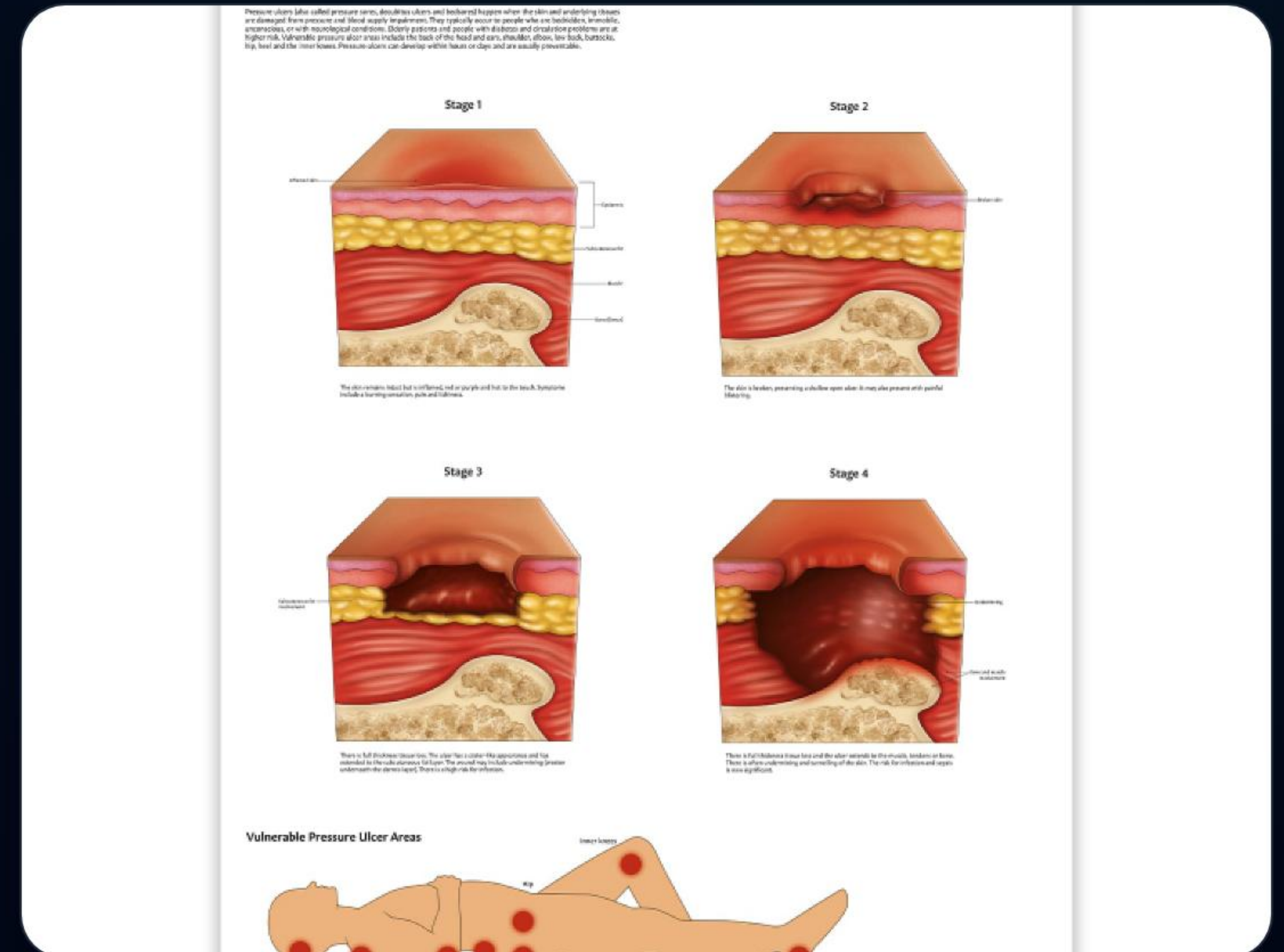
MEDICAL INTELLIGENCE UNIT

THE CLINICAL CRISIS

Every year, **millions of bedbound patients** suffer from life-threatening pressure ulcers due to prolonged ischemia.

Traditional prevention is **passive and costly**, often failing in resource-limited hospitals.

Our solution captures precise pathophysiology **at the bedside** to intervene before necrosis begins.



| THE CLINICAL RISK INDEX (RI)

$$RI = 0.50P + 0.30T + 0.20M$$

 Pressure

 Temp

 Moisture

Predictive Pathophysiology

By streaming real-time tissue markers, our engine calculates a custom **Multi-Parameter Risk Index**.

This shifts care from reactive management to **affordable, predictive prevention**, optimizing nursing workflows.

AFFORDABLE HARDWARE

A low-cost **\$20–\$45 sensor matrix** embedded beneath standard hospital mattresses.

Models live telemetry from a pressure-sensitive grid to generate a 24/7 skin health map without manual inspection.



| THE PIPELINE



Telemetry

Real-time data streaming of Pressure, Temperature, and Moisture markers.



ML Analysis

Semantic Context Engine (RAG) matches live data with clinical research findings.



Action

Instant nursing instructions generated for ward staff to optimize patient care.

Predictive Care

Turning Research into Real-Time Prevention

hf.co/spaces/bkbilal09/neurotech-bedside-rag

Hackathon 2026 | Team NeuroTech

| IMAGE SOURCES



<http://www.anatomystuff.co.uk/cdn/shop/files/stages-of-pressure-ulcers-poster-product-image.jpg?v=1741092323>

Source: www.anatomystuff.co.uk



https://www.johnpreston.co.uk/media/catalog/product/cache/0b3ab3051aff204293f33c55714d6d6c/t/a/tactilus_mattress_pressure_maps_1.jpg

Source: www.johnpreston.co.uk